



THE BETTER POLICY PROJECT

CENTRAL BANK OF ARMENIA · WORKING PAPER 2022/07



AI-POWERED LEARNING

Getting FIT with Imperfect Policy Credibility

A small-open-economy model with endogenous credibility — calibrated to Armenia

Asya Kostanyan · Anahit Matinyan · Angela Papikyan with the FPAS Mark II team, Central Bank of Armenia

01

The Model

From the closed-economy ENDOCRED to a small open economy

02

Credibility & Nonlinearities

Why trust, convexity, and a loss function change the policy answer

03

The Case of Armenia

Historical narrative, the Q1-2022 scenarios, and the folly analysis



OVERVIEW

What This Paper Develops

A nonlinear workhorse QPM for flexible-inflation-targeting small open economies



Open ENDOCRED model

A small-open-economy extension of the closed-economy ENDOCRED workhorse (Kostanyan et al. 2022b), in the spirit of Argov et al. (2007).



Credibility × country risk

Endogenous policy credibility interacts with the country-risk premium and the real exchange rate — the new external channels.



A core QPM for FPAS Mark II

Built to support monetary policy as a risk-management exercise under deep uncertainty and crucial nonlinearities.



Calibrated to Armenia

The first calibration, with empirical estimates of credibility and a real-world Q1-2022 application.



WHY THIS MODEL

Three Departures from the Standard QPM

What FPAS Mark I models leave out — and this model puts back in



01

Endogenous policy credibility

Credibility is a stock: lost quickly when inflation is high, persistent, and expectations de-anchor — but regained only slowly and nonlinearly.



02

Crucial nonlinearities

A convex, inflation-expectations-augmented Phillips curve and a nonlinear credibility-generation process.



03

Explicit loss function

Optimal policy minimizes a quadratic loss over the full horizon, replacing the conventional interest-rate reaction function.



FROM CLOSED TO OPEN

Extending ENDOCRED for a Small Open Economy

Same philosophy as the closed-economy model — with new external channels

	Closed ENDOCRED (US)	Open ENDOCRED (Armenia)
Benchmark application	United States	Armenia
Exchange rate	— (closed)	Real exchange rate; UIP with risk premium
Output-gap drivers	Rate gap, term premium	+ Real exchange-rate gap, world output gap
Inflation / Phillips	Convex in the output gap	+ Direct & indirect FX pass-through
Credibility channel	Expectations only	+ Dollarization / capital flight via the premium
Max excess demand $\hat{y}_{\max}$	5	6 (imports relax capacity)
Source	Kostanyan et al. (2022b)	This paper · Argov et al. (2007)



ENDOGENOUS CREDIBILITY

How Much Agents Trust the Bank?

**THE BIG IDEA**

Agents form expectations as a weighted mix of the target and observed past inflation. The weight — credibility — evolves with the bank's track record, so credibility itself becomes a state variable.

HIGH CREDIBILITY**Expectations anchored**

- High weight on the target
- Deviations read as transitory
- Policy can act gradually
- Shocks absorbed by the system

LOW CREDIBILITY**Expectations drifting**

- More weight on observed past inflation
- Deviations persistent & self-reinforcing
- Policy must act forcefully to regain control
- The cost of late action rises sharply



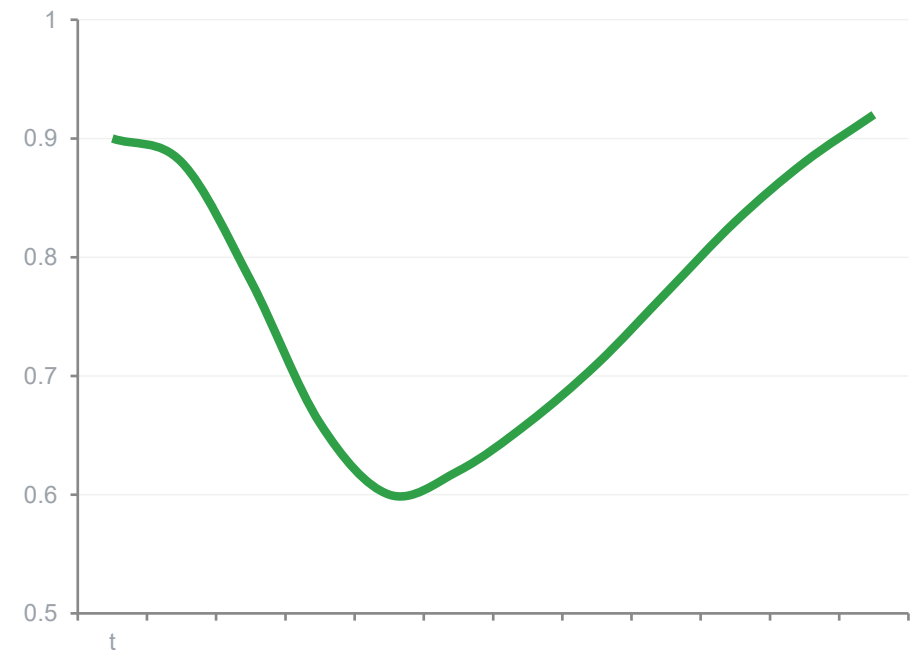
THE CREDIBILITY STOCK

Credibility: Slow to Earn, Quick to Lose

A bounded stock, updated from the bank's inflation track record

$$C_t = \omega S_t + (1 - \omega) C_{t-1} \quad S_t = \frac{e_{H,t}^2}{e_{H,t}^2 + e_{L,t}^2}$$

Illustrative credibility path

**Bounded weight**

$\omega \in [0,1]$ sets how forward- vs backward-looking expectations are.

Bayesian-style signal

S compares outcomes against a credible-target regime vs a high-inflation regime.

Asymmetric & nonlinear

Credibility erodes fast; re-anchoring is gradual.

Armenia

≈ 0.9 before the shock, falling toward ~ 0.6 under persistent inflation.



Forward- and Backward-Looking Expectations

More backward-looking than a standard DSGE model — by design

$$\pi_{t+4}^e = C_t \pi_{t+4} + (1 - C_t) \pi_{t-1} + \pi_t^b + (1 - C_t) \varepsilon_t^{z+} + \varepsilon_t^{\pi e}$$

C_t weight on the model-consistent forward forecast (credibility).

$(1 - C_t)$ weight on last year's observed inflation — the backward-looking pull.

π_t^b bias term: expectations ratchet up as credibility falls.

$(1 - C_t) \cdot \varepsilon_t^{z+}$ asymmetric pass-through — depreciation shocks bite in low-credibility states; appreciations are largely ignored.



NONLINEARITY

The Convex Phillips Curve

Where linear approximations hide the dark corners

$$\pi_t = \lambda_1 \pi_{t-4}^e + (1 - \lambda_1) \pi_{t-1} + \lambda_2 \frac{\hat{y}_{max} \hat{y}_{t-1}}{\hat{y}_{max} - \hat{y}_{t-1}} + \beta_1 (\hat{z}_t - \hat{z}_{t-1}) + \beta_2 \hat{z}_t + \varepsilon_t^\pi$$

Convexity

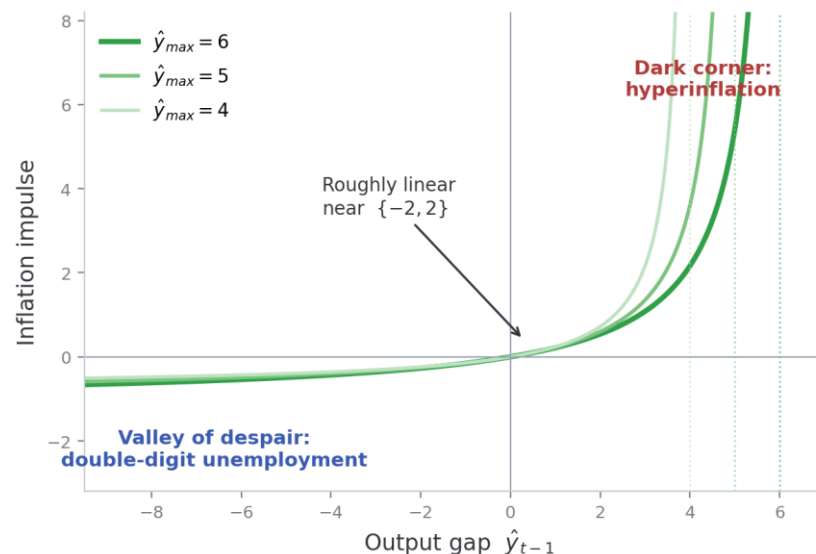
Near $\hat{y}_{max}$ extra demand is increasingly inflationary (a vertical asymptote); in deep excess supply the curve flattens — downward wage rigidity.

Dark corners

Linear models miss both tails: runaway inflation on one side, a double-digit-unemployment valley on the other.

$\hat{y}_{max} = 6$ (vs 5 closed)

An open economy can import to relax capacity constraints.





Output Gap: Now with External Demand

Two new terms open the closed-economy output gap

$$\hat{y}_t = \beta_1 \hat{y}_{t-1} + \beta_2 \hat{y}_{t+1} - \beta_3 \hat{r}_{t-1} + \beta_4 \hat{z}_{t-1} + \beta_5 \hat{y}_t^w + \varepsilon_t^{\hat{y}}$$

$\beta_1 \hat{y}_{t-1}, \beta_2 \hat{y}_{t+1}$ inertia vs forward-looking behaviour — set by how rigid the economy is.

$-\beta_3 \hat{r}_{t-1}$ the real interest-rate gap: the demand channel of policy.

$+\beta_4 \hat{z}_{t-1}$ the real exchange-rate gap: competitiveness ($\hat{z} > 0$ = depreciation).

$+\beta_5 \hat{y}^w$ the world output gap: external demand and openness.

β_1 0.50–0.70 β_2 0.10–0.25 β_3 0.15–0.40 β_4 0.00–0.20 β_5 0.12–0.24



Real Interest Parity with a Country-Risk Premium

$$r_t = (r_t^{US} + \psi_t) + (\zeta_{t+1}^e - \zeta_t) \times 4$$

$$\psi_t = \beta_1 \bar{\psi}^{**} + (1 - \beta_1) \psi_{t-1} + \beta_2 \pi 4_t^b + \beta_3 (\pi 4_t^b - \pi 4_{t-1}^b) + \varepsilon_t^\psi$$

Real parity The domestic real rate equals the US real rate, plus the risk premium, plus the expected change in the real exchange rate.

A credibility channel As credibility falls, the bias term $\pi 4^b$ rises $\rightarrow$ dollarization and capital flight $\rightarrow$ nominal depreciation (positive sign).

Equilibrium premium $\bar{\psi}^{**}$ follows a random walk; current-account and net-foreign-liability stock-flow dynamics are left to satellite models.



The Credibility Bias and the Policy Objective

Bias term in inflation expectations

$$\pi 4_t^b = \kappa \left[\gamma_t \pi 4_{t+4}^{eL} + (1 - \gamma_t) \pi 4_{t+4}^{eH} - \pi 4_t^* \right]$$

Monetary-policy loss function

$$Loss_t = \sum_{j=0}^{\infty} \rho^j \left[\omega_1 (\pi 4_{t+j} - \pi^*)^2 + \omega_2 \hat{y}_{t+j}^2 + \omega_3 (i_{t+j} - i_{t+j-1})^2 \right]$$

Bias ($\kappa = 0.10$)

Blends low- and high-credibility expectation regimes — a source of stagflationary shocks in dollarized economies.

Loss ($\rho = 0.95$, $\omega_1 = 1$)

Penalizes squared inflation-gap, output gap, and one-quarter rate changes.

Why a loss, not a rule?

Robust to structural change across scenarios, and it yields a full-horizon total-loss metric for risk assessment.



CALIBRATION

Calibration at a Glance

Ranges and central values — critical judgment is needed for each economy

Parameter	Role	Range (central)
λ_1	Forward weight in the Phillips curve	0.55 – 0.80
λ_2	Output-gap slope (convex term)	0.15 – 0.40 (0.18)
$\hat{y}_{\max}$	Maximum excess demand	5 – 7 (6)
$\beta_1, \beta_2 (\pi)$	Direct / indirect FX pass-through	0.05 – 0.25 (0.20)
κ	Expectations bias	0.00 – 0.30 (0.10)
$\beta_1 \dots \beta_5 (\hat{y})$	Output-gap weights	0.50–0.70 / 0.10–0.25 / 0.15–0.40 / 0.00–0.20 / 0.12–0.24
$\beta_2, \beta_3 (\psi)$	Credibility bias in the risk premium	0.66 / 2.30
$\beta_4 (\zeta)$	Forward-looking weight in the FX equation	0.50 – 1.00 (0.70)
$\rho, \omega_1 / \omega_2 / \omega_3$	Loss discount and weights	0.95, 1 / 0.5–1.5 / 0.4–0.6



THE CASE OF ARMENIA

Two Decades in Four Acts

Inflation targeting since 2006 — and the making (and testing) of credibility

01

2006–2009

Double-digit growth

Capital inflows, ~50% appreciation, a real-estate boom — demand-driven inflation despite a strong dram.

02

2009–2014

Slow growth, high risk premia

GFC sudden stop, debt-sustainability doubts; sticky-price inflation kept expectations elevated.

03

2014–2020

Disinflation

Ruble-shock spillover, then +12pp of tightening and a disinflation that rebuilt credibility and cut dollarization.

04

2020–2022

Post-war, post-pandemic

Early proactive hikes (Dec-2020); by late 2022 the policy rate reached 10.5% amid inflows and overheating.



Two Risk-Management Scenarios

Case A = hawkish (rate above market) **Case B** = dovish (rate below market) **Case X** = tail risks / dark corners

CASE A

Persistent inflation, demand pressures

- War creates a preference shock for Armenian goods, plus Russian migration and tourism
- Bottlenecks steepen the Phillips curve; the dram appreciates
- Policy must move fast → rate peaks $\approx 15\%$, credibility falls ~ 0.9 → ~ 0.6

CASE B

A mixed bag

- Excess capacity absorbs demand; bottlenecks hit later
- Lower remittances soften demand; a risk-premium shock brings some depreciation
- Rate still rises, but less — inflation returns with smaller hikes



What If You Get the Framework Wrong?

Re-running the Case A scenario under flawed assumptions

B.1

Delayed response

Holding rates for two quarters forces a higher peak ($\approx 18\%$ vs 15%) and a deeper cumulative output loss ($\approx -6\%$ vs -3%).

B.2

Linear + perfect credibility

A DSGE-like linear model peaks at only $\approx 12\%$ — systematically underestimating the response actually needed.

B.3

Linear + delayed

The two errors compound: the bank never sees the imperative to act — echoing the 1970s Great Inflation.

Assuming perfect credibility and linearity tricks policymakers into “too little, too late.”



What Actually Happened: Q1–Q2 2022

The realized episode behind the model's Case of Armenia

9.25%

Refinancing rate, held in June 2022 — after +3.75pp over the prior year

9.0%

12-month inflation, May 2022 (core 8.4%) — above the 4% target

4.9%

2022 growth, revised up from 1.6% in the previous program

+8.6%

Q1 2022 GDP — services- and demand-led

+8.85%

Dram vs USD, Mar–May (real effective +4.88%)

H2 2023

Projected return toward the 4% inflation target

Source: Central Bank of Armenia, Inflation Report 2022/2 (Monetary Policy Program Q2 2022; Status Report Q1 2022).



WHAT DROVE IT

The Inflow Shock, in the Data

A demand and capital inflow — predominantly from Russia — that appreciated the dram

Visitor mix shifted

Russian citizens were 43.7% of Q1 tourists, up from 34.9% a year earlier.

Services surged

Accommodation, catering, apartment rents and financial services rose sharply.

Capital and labour arrived

New foreign-founded firms — mainly IT — and a jump in non-resident bank accounts.

The dram as shock absorber

Under a floating regime the currency softens the shock (Friedman, 1953).

WHAT IT MEANT FOR INFLATION

≈ 1.2 pp

of 12-month inflation offset by the appreciation
(Apr–May 2022)

4.9% vs 3.6%

2022 growth with vs without the inflows (Bank's counterfactual) — without them, lower short-run inflation and less appreciation.



POLICY RESPONSE

The Bank's Read — and the Decision

THE DILEMMA • INFLATION REPORT, BOX 3

FX absorbs the shock

Under floating rates the real exchange rate appreciates either way — the currency does much of the adjustment.

Policy manages the residual

Appreciation eases imported inflation; monetary policy handles the residual de-anchoring risk.

Fixing shifts the burden

Intervening to hold the rate pushes adjustment onto domestic inflation and the policy rate.

Sectoral risk ≠ monetary tool

“Dutch-disease” risks in services are best met with targeted fiscal policy.

THE JUNE 2022 DECISION

- Rate held at 9.25% — prior tightening judged sufficient
- Inflation seen returning toward 4% from H2 2023
- Risks to the outlook assessed as balanced
- Risk premium rose, stabilized, then rose again with global tightening — long-run level unchanged

Case A, realized: a demand-driven appreciation, a credibility-protecting hold, and the exchange rate doing much of the adjustment.



From Expert Forecaster to Risk Manager

KEY TAKEAWAYS

Credibility is endogenous

Treating it as given underestimates the timing and size of the needed response.

Nonlinearities matter

Convexity and the credibility process create real welfare costs that linear models hide.

The tool needs the framework

ENDOCRED adds value only inside an FPAS Mark II, scenario-based, transparent-communication approach.

The goal

Avoid the “dark corners where danger lurks.”



DISCUSSION QUESTIONS

1. What would scenario-based (A/B/X) communication change at your central bank?
2. How would you measure credibility in your economy — what data would you reach for?
3. Where would endogenous credibility most have changed a past policy call?
4. What governance is needed before such a model is decision-grade?